PV102 Day 6: Inverter Charge Controller and Protection Boundaries

Students map the balance-of-system decisions that make PV energy usable and controlled: charge controllers, inverters, disconnects, overcurrent protection, grounding/bonding boundaries, and labeled handoff points.

PV102 | CTS Day 22 | Instructor Run Sheet
Power electronics set system boundaries. | Tuesday, August 11, 2026

Core rule

Power electronics set system boundaries.

Learning targets

- Explain the roles of charge controllers and inverters in battery-based PV systems.
- Match inverter selection to continuous load, surge load, voltage, and output requirement.
- Identify where disconnects and overcurrent protection belong conceptually.
- Create a block diagram that separates source, storage, conversion, and load boundaries.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Component cards: PV, controller, battery, inverter, OCPD, disconnect, load String/yarn or arrows Calculators Sample labels Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 5 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review

		acceptance criteria and stop-work triggers.
1:00-2:15	Balance-of-System Block Diagram	Teams build a labeled block diagram and defend every protection/isolation point.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 7 turns the design into an install plan, tool/material list, JHA, and daily work sequence.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use inverter charge controller and protection boundaries to make the next design decision safer and more defensible. The rule is simple: Power electronics set system boundaries."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Choose a scenario and place component cards in energy-path order.
- Draw source, storage, conversion, and load boundaries.
- Add conceptual OCPD/disconnect labels where energy could feed a fault or where service isolation is needed.
- Mark manufacturer-instruction-required decisions.
- Run a peer safety check: can someone tell what is energized, isolated, and unknown?
- Handoff the diagram and unresolved decisions.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 7 turns the design into an install plan, tool/material list, JHA, and daily work sequence.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 6 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 22 | Morning Quiz - Answer Key
Power electronics set system boundaries. | Tuesday, August 11, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. An inverter primarily converts:

- A. DC to AC
- B. AC to sunlight
- C. wood to wire
- D. shade to volts

Answer: A

Rationale/look-for: Inverters convert DC energy to AC output for AC loads.

2. A charge controller protects and manages:

- A. Battery charging from PV
- B. Roof pitch
- C. Clipboards
- D. Nameplate stickers only

Answer: A

Rationale/look-for: Controllers regulate charging within battery and PV limits.

3. Inverter sizing must consider:

- A. Continuous load and surge
- B. Marker color only
- C. Roof direction only
- D. Lunch menu

Answer: A

Rationale/look-for: Continuous and starting/surge requirements both matter.

4. An overcurrent device is used to:

- A. Open a circuit during excessive current
- B. Store solar energy
- C. Increase sunshine
- D. Convert AC to shade

Answer: A

Rationale/look-for: OCPDs limit damaging current when properly selected and installed.

5. A disconnect provides:

- A. A means to isolate equipment
- B. Free battery capacity
- C. More roof area
- D. A better compass

Answer: A

Rationale/look-for: Disconnects support isolation for service and emergency boundaries.

6. Why should source, storage, conversion, and load boundaries be shown separately?

_____

_____

Answer: Clarity/safety

Rationale/look-for: It clarifies what each component controls and where hazards/isolations exist.

7. Name one label that belongs on a training system handoff diagram.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept voltage, current, source, battery, disconnect, OCPD rating, load panel, do not energize, polarity, or owner/equipment ID.

8. Why can an inverter with enough continuous watts still fail a job?

_____

_____

Answer: Surge/output/voltage

Rationale/look-for: Surge, output voltage, waveform, DC input, or environmental limits may not match.

9. What is one thing students should not decide without manufacturer instructions?

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept torque, wiring sequence, fuse size, grounding detail, settings, charge profile.

10. Write the Day 6 rule in your own words.

_____

_____

Answer: Power electronics boundaries

Rationale/look-for: Look for controller/inverter as hard limits and conversion boundaries.

PV102 Day 6 Whiteboard Outline

PV102 | CTS Day 22 | Whiteboard Outline
Power electronics set system boundaries. | Tuesday, August 11, 2026

Board 1: Job	Board 2: Evidence	Board 3: Handoff
Topic: Inverter Charge Controller and Protection Boundaries Rule: Power electronics set system boundaries. Question: What job are we solving?	Measured / Observed / Unknown Constraint that matters most Safety boundary before work	Decision Evidence Risk Unknowns Next action

Scenario anchors

- 12 V training system with PV module, controller, LiFePO4 battery, inverter, DC load, AC load.
- 48 V bus system concept with hybrid inverter/charger, battery bank, PV strings, AC distribution.
- Remote DC-only sensor system with no inverter and limited service visits.

Final board sentence

Power electronics set system boundaries.

PV102 Day 6 Student Packet

Inverter Charge Controller and Protection Boundaries

PV102 | CTS Day 22 | Student Print Packet
Power electronics set system boundaries. | Tuesday, August 11, 2026

Today's rule

Power electronics set system boundaries.

Learning targets

- Explain the roles of charge controllers and inverters in battery-based PV systems.
- Match inverter selection to continuous load, surge load, voltage, and output requirement.
- Identify where disconnects and overcurrent protection belong conceptually.
- Create a block diagram that separates source, storage, conversion, and load boundaries.

Morning quiz

1. An inverter primarily converts:

- A. DC to AC
- B. AC to sunlight
- C. wood to wire
- D. shade to volts

2. A charge controller protects and manages:

- A. Battery charging from PV
- B. Roof pitch
- C. Clipboards
- D. Nameplate stickers only

3. Inverter sizing must consider:

- A. Continuous load and surge
- B. Marker color only
- C. Roof direction only
- D. Lunch menu

4. An overcurrent device is used to:

- A. Open a circuit during excessive current
- B. Store solar energy
- C. Increase sunshine
- D. Convert AC to shade

5. A disconnect provides:

- A. A means to isolate equipment
- B. Free battery capacity
- C. More roof area
- D. A better compass

6. Why should source, storage, conversion, and load boundaries be shown separately?

_____

_____

7. Name one label that belongs on a training system handoff diagram.

_____

_____

8. Why can an inverter with enough continuous watts still fail a job?

_____

_____

9. What is one thing students should not decide without manufacturer instructions?

_____

_____

10. Write the Day 6 rule in your own words.

_____

_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Balance-of-System Block Diagram

Teams build a labeled block diagram and defend every protection/isolation point.

Scenario options

- 12 V training system with PV module, controller, LiFePO4 battery, inverter, DC load, AC load.
- 48 V bus system concept with hybrid inverter/charger, battery bank, PV strings, AC distribution.
- Remote DC-only sensor system with no inverter and limited service visits.

Team procedure

- Choose a scenario and place component cards in energy-path order.
- Draw source, storage, conversion, and load boundaries.
- Add conceptual OCPD/disconnect labels where energy could feed a fault or where service isolation is needed.
- Mark manufacturer-instruction-required decisions.
- Run a peer safety check: can someone tell what is energized, isolated, and unknown?
- Handoff the diagram and unresolved decisions.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which boundary on your diagram mattered most for safety, and what would happen if it were missing?

_____

_____

PV102 Day 6 Afternoon Lab

Balance-of-System Block Diagram

PV102 | CTS Day 22 | Afternoon Lab
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Mission

Teams build a labeled block diagram and defend every protection/isolation point.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- 12 V training system with PV module, controller, LiFePO4 battery, inverter, DC load, AC load.
- 48 V bus system concept with hybrid inverter/charger, battery bank, PV strings, AC distribution.
- Remote DC-only sensor system with no inverter and limited service visits.

Procedure

- Choose a scenario and place component cards in energy-path order.
- Draw source, storage, conversion, and load boundaries.
- Add conceptual OCPD/disconnect labels where energy could feed a fault or where service isolation is needed.
- Mark manufacturer-instruction-required decisions.
- Run a peer safety check: can someone tell what is energized, isolated, and unknown?
- Handoff the diagram and unresolved decisions.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	

Primary decision	
Evidence	
Safety boundary	
Unknowns	
Next safest action	

PV102 Day 6 Classroom Post

Copy-ready announcement and assignment text

PV102 | CTS Day 22 | Google Classroom Post
Power electronics set system boundaries. | Tuesday, August 11, 2026

Announcement

Today in PV102 we are working on inverter charge controller and protection boundaries. The core rule is: Power electronics set system boundaries.

Assignment title

PV102 Day 6 - Inverter Charge Controller and Protection Boundaries

Student directions

- Complete the morning readiness quiz independently.
- Use the reference section during the afternoon lab.
- Work with your team to complete the scenario handoff.
- Complete the exit ticket before class ends.

Turn in

- Completed student packet
- Team lab handoff
- Exit ticket

Success criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record unknowns.